

FEATURES

High performance

- Pseudo differential analog input range
0 V to V_{REF} with V_{REF} between 2.5 V to 5 V
- Throughput: 1 MSPS
- Zero latency architecture
- 16-bit resolution with no missing codes
- INL: ± 0.6 LSB typical, ± 1.25 LSB maximum
- Dynamic range: 92 dB, $V_{REF} = 5$ V
- SNR: 91.5 dB at $f_{IN} = 10$ kHz, $V_{REF} = 5$ V
- THD: -114 dB at $f_{IN} = 10$ kHz, $V_{REF} = 5$ V
- SINAD: 91 dB at $f_{IN} = 10$ kHz, $V_{REF} = 5$ V

Low power dissipation

- Single-supply 2.5 V operation with 1.8 V/2.5 V/3 V/5 V logic interface
- 4 mW at 1 MSPS (V_{DD} only)
- 7 mW at 1 MSPS (total)
- 70 μ W at 10 kSPS

Proprietary serial interface

- SPI/QSPI/MICROWIRE™/DSP compatible
- Daisy-chain multiple ADCs and busy indicator
- 10-lead MSOP, 10-lead, 3 mm $\times$ 3 mm LFCSP
- Wide operating temperature range: -40°C to $+125^{\circ}\text{C}$

APPLICATIONS

- Automated test equipment
- Data acquisition systems
- Medical instruments
- Machine automation

TYPICAL APPLICATION CIRCUIT

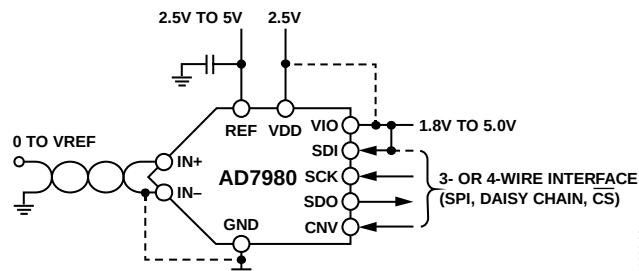


Figure 1.

GENERAL DESCRIPTION

The **AD7980**¹ is a 16-bit, successive approximation, analog-to-digital converter (ADC) that operates from a single power supply, V_{DD} . It contains a low power, high speed, 16-bit sampling ADC and a versatile serial interface port. On the CNV rising edge, it samples an analog input, $IN+$, between 0 V to REF with respect to a ground sense, $IN-$. The reference voltage, REF , is applied externally and can be set independent of the supply voltage, V_{DD} . Its power scales linearly with throughput.

The SPI-compatible serial interface also features the ability, using the SDI input, to daisy-chain several ADCs on a single, 3-wire bus and provides an optional busy indicator. It is compatible with 1.8 V, 2.5 V, 3 V, or 5 V logic, using the separate supply VIO .

The **AD7980** is housed in a 10-lead MSOP or a 10-lead LFCSP with operation specified from -40°C to $+125^{\circ}\text{C}$.

¹ Protected by U.S. Patent 6,703,961.

Table 1. MSOP, LFCSP 16-/18-/20-Bit Precision SAR ADCs and SAR ADC-Based μ Module Data Acquisition Solutions

Type	≤ 100 kSPS	≤ 250 kSPS	≤ 500 kSPS	≤ 1000 kSPS	≤ 2000 kSPS	μ Module Data Acquisition Solutions
Differential						
20-Bit			AD4022 ¹	AD4021 ¹	AD4020 ¹	
18-Bit	AD7989-1 ¹	AD7691 ¹	AD4011 ¹ AD7690 ¹ AD7989-5 ¹	AD4007 ¹ AD7982 ¹ AD7984 ¹	AD4003 ¹	
16-Bit	AD7684	AD7687 ¹	AD7688 ¹ AD7693 ¹ AD7916 ¹	AD4005 ¹ AD7915 ¹	AD4001 ¹	
Pseudo Differential						
18-Bit			AD4010 ¹	AD4006 ¹	AD4002 ¹	
16-Bit	AD7988-1 ¹ AD7680 AD7683	AD7685 ¹ AD7694	AD4008 ¹ AD7988-5 ¹ AD7686 ¹	AD4004 ¹ AD7980 ¹ AD7983 ¹	AD4000 ¹	ADAQ7980 ADAQ7988

¹ Pin for pin compatible.

Rev. G

Document Feedback

Information furnished by Analog Devices is believed to be accurate and reliable. However, no responsibility is assumed by Analog Devices for its use, nor for any infringements of patents or other rights of third parties that may result from its use. Specifications subject to change without notice. No license is granted by implication or otherwise under any patent or patent rights of Analog Devices. Trademarks and registered trademarks are the property of their respective owners.

TABLE OF CONTENTS

Features	1
Applications	1
Typical Application Circuit	1
General Description	1
Revision History	2
Specifications	4
Timing Specifications	6
Absolute Maximum Ratings	8
Thermal Resistance	8
ESD Caution.....	8
Pin Configurations and Function Descriptions.....	9
Terminology.....	10
Typical Performance Characteristics.....	11
Theory of Operation	15
Circuit Information	15
Converter Operation.....	15
Typical Application Circuit with Multiple Supplies	16

Analog Input.....	17
Driver Amplifier Choice	17
Voltage Reference Input.....	18
Power Supply	18
Digital Interface.....	18
3-Wire $\overline{\text{CS}}$ Mode Without Busy Indicator.....	19
3-Wire $\overline{\text{CS}}$ Mode with Busy Indicator.....	20
4-Wire $\overline{\text{CS}}$ Mode Without Busy Indicator.....	21
4-Wire $\overline{\text{CS}}$ Mode with Busy Indicator.....	22
Chain Mode Without Busy Indicator	23
Chain Mode with Busy Indicator.....	24
Applications Information	25
Layout	25
Evaluating the Performance of the AD7980.....	25
Outline Dimensions.....	26
Ordering Guide	27

REVISION HISTORY

1/2021—Rev. F to Rev. G

Changes to Features Section, Applications Section, and Table 1.....	1
Changes to Table 2	4
Changes to Timing Specifications Section and Table 4.....	6
Added Endnote 1 and Endnote 2, Table 5.....	7
Deleted Figure 3; Renumbered Sequentially	7
Changes to Table 6.....	8
Added Thermal Resistance Section and Table 7; Renumbered Sequentially	8
Changes to Figure 26.....	16
Changes to Driver Amplifier Choice Section and Table 10	17
Changes to Voltage Reference Input Section and Power Supply Section.....	18

10/2017—Rev. E to Rev. F

Changes to Table 1	1
Changes to Figure 5.....	8
Updated Outline Dimensions.....	26
Changes to Ordering Guide.....	26

7/2016—Rev. D to Rev. E

Changed VIO = 2.3 V to 5.5 V to VIO = 1.71 V to 5.5 V	Throughout
Change to Features Section.....	1
Changes to Conversion Rate Parameter, Table 2	3
Changes to VIO Parameter, Table 3.....	4
Deleted VIO Range Parameter, Table 3	4
Added Table 5; Renumbered Sequentially	6
Changes to Table 7	8
Changes to Table 9.....	16
Changes to Voltage Reference Input Section	17
Changes to Figure 32	18
Changes to Figure 34	19
Changes to Figure 36	20
Changes to Figure 38	21
Changes to Figure 40	22
Changes to Figure 42	23

7/2014—Rev. C to Rev. D

Changed QFN (LFCSP) to LFCSP	Throughout
Changes to Features Section and Table 1	1
Added Patent Note, Note 1	1
Changes to AC Accuracy Parameter, Table 2	3
Change to Standby Current Parameter, Table 3	4
Changes to Figure 25	13
Changes to Table 8	15
Changes to Power Supply Section	16

8/2013—Rev. B to Rev. C

Change to Features Section	1
Changes to Table 3	4
Change to Figure 5	7
Added EPAD Row, Table 6	7

Changes to Evaluating the Performance of the AD7980 Section	23
Updated Outline Dimensions	24
Changes to Ordering Guide	25

6/2009—Rev. A to Rev. B

Changes to Table 5	6
Changes to Figure 25	13
Updated Outline Dimensions	24
Changes to Ordering Guide	25

9/2008—Rev. 0 to Rev. A

Deleted QFN Endnote	Throughout
Changes to Ordering Guide	24

8/2007—Revision 0: Initial Version

SPECIFICATIONS

VDD = 2.5 V, VIO = 1.71 V to 5.5 V, VREF = 5 V, TA = -40°C to +125°C, unless otherwise noted.

Table 2.

Parameter	Test Conditions/Comments	A Grade			B Grade			Unit
		Min	Typ	Max	Min	Typ	Max	
RESOLUTION		16			16			Bits
ANALOG INPUT								
Voltage Range	IN+ – IN–	0		VREF	0		VREF	V
Absolute Input Voltage	IN+	–0.1		VREF + 0.1	–0.1		VREF + 0.1	V
	IN–	–0.1		+0.1	–0.1		+0.1	V
Analog Input CMRR	fIN = 100 kHz		60			60		dB
Leakage Current at 25°C	Acquisition phase		1			1		nA
Input Impedance			See the Analog Input section			See the Analog Input section		
ACCURACY								
No Missing Codes		16			16			Bits
Differential Linearity Error	REF = 5 V	–1.0	±0.5	+2.0	–0.9	±0.4	+0.9	LSB ¹
	REF = 2.5 V		±0.7			±0.55		LSB ¹
Integral Linearity Error	REF = 5 V	–2.5	±1.5	+2.5	–1.25	±0.6	+1.25	LSB ¹
	REF = 2.5 V		±1.65			±0.65		LSB ¹
Transition Noise	REF = 5 V		0.75			0.6		LSB ¹
	REF = 2.5 V		1.2			1.0		LSB ¹
Gain Error, TMIN to TMAX ²			±2			±2		LSB ¹
Gain Error Temperature Drift			±0.35			±0.35		ppm/°C
Zero Error, TMIN to TMAX ²		–1.0	±0.08	+1.0	–0.5	±0.08	+0.5	mV
Zero Temperature Drift			0.54			0.54		ppm/°C
Power Supply Sensitivity	VDD = 2.5 V ± 5%		±0.1			±0.1		LSB ¹
THROUGHPUT								
Conversion Rate	VIO ≥ 3.3 V	0		1	0		1	MSPS
	VIO < 3.3 V, 85°C to 125°C			833			833	kSPS
	VIO > 2.3 V, up to 85°C			1			1	MSPS
	VIO ≤ 2.3 V			833			833	kSPS
Transient Response	Full-scale step			290			290	ns
AC ACCURACY								
Dynamic Range	VREF = 5 V		91			92		dB ³
	VREF = 2.5 V		86			87		dB ³
Oversampled Dynamic Range	fO = 10 kSPS		110			111		dB ³
Signal-to-Noise Ratio, SNR	fIN = 10 kHz, VREF = 5 V		90.5		90	91.5		dB ³
	fIN = 10 kHz, VREF = 2.5 V		86.0			87.0		dB ³
Spurious-Free Dynamic Range, SFDR	fIN = 10 kHz		–103.5			–110		dB ³
Total Harmonic Distortion, THD	fIN = 10 kHz		–101			–114		dB ³
Signal-to-Noise-and-Distortion Ratio, SINAD	fIN = 10 kHz, VREF = 5 V		90			91		dB ³
	fIN = 10 kHz, VREF = 2.5 V		85.5			86.5		dB ³

¹ LSB means least significant bit. With the 5 V input range, 1 LSB is 76.3 μV.

² See the Terminology section. These specifications include full temperature range variation, but not the error contribution from the external reference.

³ All specifications in dB are referred to a full-scale range (FSR). Tested with an input signal at 0.5 dB below full scale, unless otherwise specified.

VDD = 2.5 V, VIO = 1.71 V to 5.5 V, V_{REF} = 5 V, T_A = -40°C to +125°C, unless otherwise noted.

Table 3.

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
REFERENCE					
Voltage Range		2.4		5.1	V
Load Current	1 MSPS, REF = 5 V		330		μA
SAMPLING DYNAMICS					
-3 dB Input Bandwidth			10		MHz
Aperture Delay	VDD = 2.5 V		2.0		ns
DIGITAL INPUTS					
Logic Levels					
V _{IL}	VIO > 3V	-0.3		0.3 × VIO	V
V _{IH}	VIO > 3V	0.7 × VIO		VIO + 0.3	V
V _{IL}	VIO ≤ 3V	-0.3		0.1 × VIO	V
V _{IH}	VIO ≤ 3V	0.9 × VIO		VIO + 0.3	μA
I _{IL}		-1		+1	μA
I _{IH}		-1		+1	μA
DIGITAL OUTPUTS					
Data Format		Serial 16 bits straight binary			
Pipeline Delay		Conversion results available immediately after completed conversion			
V _{OL}	I _{SINK} = 500 μA			0.4	V
V _{OH}	I _{SOURCE} = -500 μA	VIO - 0.3			V
POWER SUPPLIES					
VDD		2.375	2.5	2.625	V
VIO		1.71		5.5	V
Standby Current ^{1, 2}	VDD and VIO = 2.5 V, 25°C		0.35		μA
Power Dissipation	VDD = 2.625 V, V _{REF} = 5 V, VIO = 3 V				
Total	10 kSPS throughput		70		μW
	1 MSPS throughput, B grade		7.0	9.0	mW
	1 MSPS throughput, A grade		7.0	10	mW
VDD Only			4		mW
REF Only			1.7		mW
VIO Only			1.3		mW
Energy per Conversion			7.0		nJ/sample
TEMPERATURE RANGE ³					
Specified Performance	T _{MIN} to T _{MAX}	-40		+125	°C

¹ With all digital inputs forced to VIO or GND as required.

² During the acquisition phase.

³ Contact sales for extended temperature range.

TIMING SPECIFICATIONS

VDD = 2.37 V to 2.63 V, VIO = 2.3 V to 5.5 V, TA = -40°C to +125°C, unless otherwise stated. See Figure 2 for load conditions.

Table 4.

Parameter ¹	Symbol	Min	Typ	Max	Unit
Throughput Rate					
VIO Above 3.3 V				1	MSPS
VIO Above 2.3 V, TA = -40°C to 85°C				1	MSPS
VIO Above 2.3 V, TA > 85°C				833	kSPS
Conversion Time: CNV Rising Edge to Data Available	tCONV				
VIO Above 3.3 V		500		710	ns
VIO Above 2.3 V, TA = -40°C to 85°C		500		710	ns
VIO Above 2.3 V, TA > 85°C		500		800	ns
Acquisition Time	tACQ	290			ns
Time Between Conversions	tCYC				
VIO Above 3.3 V		1000			ns
VIO Above 2.3 V, TA = -40°C to 85°C		1000			ns
VIO Above 2.3 V, TA > 85°C		1.2			μs
CNV Pulse Width (CS Mode)	tCNVH	10			ns
SCK Period (CS Mode)	tSCK				ns
VIO Above 4.5 V		10.5			ns
VIO Above 3 V		12			ns
VIO Above 2.7 V		13			ns
VIO Above 2.3 V		15			ns
SCK Period (Chain Mode)	tSCK				ns
VIO Above 4.5 V		11.5			ns
VIO Above 3 V		13			ns
VIO Above 2.7 V		14			ns
VIO Above 2.3 V		16			ns
SCK Low Time	tSCKL	4.5			ns
SCK High Time	tSCKH	4.5			ns
SCK Falling Edge to Data Remains Valid	tHSDO	3			ns
SCK Falling Edge to Data Valid Delay	tDSDO				
VIO Above 4.5 V				9.5	ns
VIO Above 3 V				11	ns
VIO Above 2.7 V				12	ns
VIO Above 2.3 V				14	ns
CNV or SDI Low to SDO D15 MSB Valid (CS Mode)	tEN				
VIO Above 3 V				10	ns
VIO Above 2.3 V				15	ns
CNV or SDI High or Last SCK Falling Edge to SDO High Impedance (CS Mode)	tDIS			20	ns
SDI Valid Setup Time from CNV Rising Edge	tSSDICNV	5			ns
SDI Valid Hold Time from CNV Rising Edge (CS Mode)	tHSDICNV	2			ns
SDI Valid Hold Time from CNV Rising Edge (Chain Mode)	tHSDICNV	0			ns
SCK Valid Setup Time from CNV Rising Edge (Chain Mode)	tSSCKCNV	5			ns
SCK Valid Hold Time from CNV Rising Edge (Chain Mode)	tHSCCKNV	5			ns
SDI Valid Setup Time from SCK Falling Edge (Chain Mode)	tSSDISCK	2			ns
SDI Valid Hold Time from SCK Falling Edge (Chain Mode)	tHSDISCK	3			ns
SDI High to SDO High (Chain Mode with Busy Indicator)	tDSDOSDI			15	ns

¹ Timing parameters measured with respect to a falling edge are defined as triggered at x% VIO. Timing parameters measured with respect to a rising edge are defined as triggered at y% VIO. For VIO ≤ 3 V, x = 90 and y = 10. For VIO > 3 V, x = 70 and y = 30. The minimum VIH and maximum VIL are used. See the Digital Inputs Specifications in Table 2.

VDD = 2.37 V to 2.63 V, VIO = 1.71 V to 2.3 V, T_A = -40°C to +125°C, unless otherwise stated. See Figure 2 for load conditions.

Table 5.

Parameter ¹	Symbol	Min	Typ	Max	Unit
Throughput Rate				833	kSPS
Conversion Time: CNV Rising Edge to Data Available	t _{CONV}	500		800	ns
Acquisition Time	t _{ACQ}	290			ns
Time Between Conversions ²	t _{CYC}	1.2			μs
CNV Pulse Width ($\overline{\text{CS}}$ Mode)	t _{CNVH}	10			ns
SCK Period ($\overline{\text{CS}}$ Mode)	t _{SCK}	22			ns
SCK Period (Chain Mode)	t _{SCK}	23			ns
SCK Low Time	t _{SCKL}	6			ns
SCK High Time	t _{SCKH}	6			ns
SCK Falling Edge to Data Remains Valid	t _{HSDO}	3			ns
SCK Falling Edge to Data Valid Delay	t _{DSDO}		14	21	ns
CNV or SDI Low to SDO D15 MSB Valid ($\overline{\text{CS}}$ Mode)	t _{EN}		18	40	ns
CNV or SDI High or Last SCK Falling Edge to SDO High Impedance ($\overline{\text{CS}}$ Mode)	t _{DIS}			20	ns
SDI Valid Setup Time from CNV Rising Edge	t _{SSDICNV}	5			ns
SDI Valid Hold Time from CNV Rising Edge ($\overline{\text{CS}}$ Mode)	t _{HSDICNV}	10			ns
SDI Valid Hold Time from CNV Rising Edge (Chain Mode)	t _{HSDICNV}	0			ns
SCK Valid Setup Time from CNV Rising Edge (Chain Mode)	t _{SSCKCNV}	5			ns
SCK Valid Hold Time from CNV Rising Edge (Chain Mode)	t _{HSCCKCNV}	5			ns
SDI Valid Setup Time from SCK Falling Edge (Chain Mode)	t _{SSDISCK}	2			ns
SDI Valid Hold Time from SCK Falling Edge (Chain Mode)	t _{HSDISCK}	3			ns
SDI High to SDO High (Chain Mode with Busy Indicator)	t _{DSDOSDI}			22	ns

¹ Timing parameters measured with respect to a falling edge are defined as triggered at x% VIO. Timing parameters measured with respect to a rising edge are defined as triggered at y% VIO. For VIO ≤ 3 V, x = 90 and y = 10. For VIO > 3 V, x = 70 and y = 30. The minimum V_{IH} and maximum V_{IL} are used. See the Digital Inputs Specifications in Table 2.

² The time required to clock out N bits of data, t_{READ}, may be greater than t_{ACQ}, depending on the magnitude of VIO. If t_{READ} is greater than t_{ACQ}, the throughput must be limited to ensure that all N bits are read back from the device.

Timing Diagrams

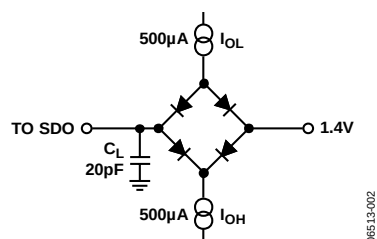


Figure 2. Load Circuit for Digital Interface Timing

ABSOLUTE MAXIMUM RATINGS

Table 6.

Parameter	Rating
Analog Inputs IN+, ¹ IN– ¹ to GND	–0.3 V to $V_{REF} + 0.3$ V or ± 130 mA
Supply Voltage REF, VIO to GND	–0.3 V to +6 V
VDD to GND	–0.3 V to +3 V
VDD to VIO	+3 V to –6 V
Digital Inputs to GND	–0.3 V to VIO + 0.3 V
Digital Outputs to GND	–0.3 V to VIO + 0.3 V
Storage Temperature Range	–65°C to +150°C
Junction Temperature	150°C
Lead Temperature	
Vapor Phase (60 sec)	215°C
Infrared (15 sec)	220°C

¹ See the Analog Input section.

Stresses at or above those listed under Absolute Maximum Ratings may cause permanent damage to the product. This is a stress rating only; functional operation of the product at these or any other conditions above those indicated in the operational section of this specification is not implied. Operation beyond the maximum operating conditions for extended periods may affect product reliability.

THERMAL RESISTANCE

Thermal performance is directly linked to printed circuit board (PCB) design and operating environment. Careful attention to PCB thermal design is required.

θ_{JA} is the natural convection junction to ambient thermal resistance measured in a one cubic foot sealed enclosure.

θ_{JC} is the junction to case thermal resistance.

Table 7. Thermal Resistance

Package Type ¹	θ_{JA}	θ_{JC}	Unit
RM-10	200	44	°C/W

¹ Test Condition 1: thermal impedance simulated values are based on use of a 252P JEDEC PCB. See the Ordering Guide.

ESD CAUTION



ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

PIN CONFIGURATIONS AND FUNCTION DESCRIPTIONS

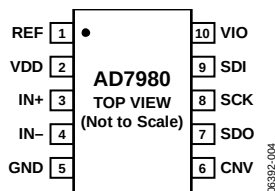
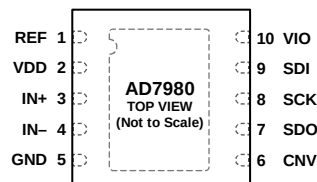


Figure 3. 10-Lead MSOP Pin Configuration



NOTES

1. CONNECT THE EXPOSED PAD TO GND. THIS CONNECTION IS NOT REQUIRED TO MEET THE ELECTRICAL PERFORMANCES.

Figure 4. 10-Lead LFCSP Pin Configuration

Table 8. Pin Function Descriptions

Pin No.		Mnemonic	Type ¹	Description
MSOP	LFCSP			
1	1	REF	AI	Reference Input Voltage. The REF range is from 2.4 V to 5.1 V. It is referred to the GND pin. This pin should be decoupled closely to the pin with a 10 μ F capacitor.
2	2	VDD	P	Power Supply.
3	3	IN+	AI	Analog Input. It is referred to IN-. The voltage range, for example, the difference between IN+ and IN-, is 0 V to V_{REF} .
4	4	IN-	AI	Analog Input Ground Sense. To be connected to the analog ground plane or to a remote sense ground.
5	5	GND	P	Power Supply Ground.
6	6	CNV	DI	Convert Input. This input has multiple functions. On its leading edge, it initiates the conversions and selects the interface mode of the device, chain, or $\overline{CS}$ mode. In $\overline{CS}$ mode, it enables the SDO pin when low. In chain mode, the data should be read when CNV is high.
7	7	SDO	DO	Serial Data Output. The conversion result is output on this pin. It is synchronized to SCK.
8	8	SCK	DI	Serial Data Clock Input. When the device is selected, the conversion result is shifted out by this clock.
9	9	SDI	DI	Serial Data Input. This input provides multiple features. It selects the interface mode of the ADC as follows. Chain mode is selected if SDI is low during the CNV rising edge. In this mode, SDI is used as a data input to daisy-chain the conversion results of two or more ADCs onto a single SDO line. The digital data level on SDI is output on SDO with a delay of 16 SCK cycles. $\overline{CS}$ mode is selected if SDI is high during the CNV rising edge. In this mode, either SDI or CNV can enable the serial output signals when low; if SDI or CNV is low when the conversion is complete, the busy indicator feature is enabled.
10	10	VIO	P	Input/Output Interface Digital Power. Nominally at the same supply as the host interface (1.8 V, 2.5 V, 3 V, or 5 V).
Not applicable	0	EPAD	Not applicable	Exposed Pad. Connect the exposed pad to GND. This connection is not required to meet the electrical performances.

¹AI = analog input, DI = digital input, DO = digital output, and P = power.

TERMINOLOGY

Integral Nonlinearity Error (INL)

INL refers to the deviation of each individual code from a line drawn from negative full scale through positive full scale. The point used as negative full scale occurs $\frac{1}{2}$ LSB before the first code transition. Positive full scale is defined as a level $1\frac{1}{2}$ LSB beyond the last code transition. The deviation is measured from the middle of each code to the true straight line (see Figure 25).

Differential Nonlinearity Error (DNL)

In an ideal ADC, code transitions are 1 LSB apart. DNL is the maximum deviation from this ideal value. It is often specified in terms of resolution for which no missing codes are guaranteed.

Offset Error

The first transition should occur at a level $\frac{1}{2}$ LSB above analog ground (38.1 μ V for the 0 V to 5 V range). The offset error is the deviation of the actual transition from that point.

Gain Error

The last transition (from 111 ... 10 to 111 ... 11) should occur for an analog voltage $1\frac{1}{2}$ LSB below the nominal full scale (4.999886 V for the 0 V to 5 V range). The gain error is the deviation of the actual level of the last transition from the ideal level after the offset is adjusted out.

Spurious-Free Dynamic Range (SFDR)

SFDR is the difference, in decibels (dB), between the rms amplitude of the input signal and the peak spurious signal.

Effective Number of Bits (ENOB)

ENOB is a measurement of the resolution with a sine wave input. It is expressed in bits and related to SINAD by the following formula:

$$ENOB = (SINAD_{dB} - 1.76)/6.02$$

Noise-Free Code Resolution

Noise-free code resolution is the number of bits beyond which it is impossible to distinctly resolve individual codes. It is calculated as

$$Noise\text{-}Free\ Code\ Resolution = \log_2(2^N/Peak\text{-}to\text{-}Peak\ Noise)$$

and is expressed in bits.

Effective Resolution

Effective resolution is calculated as

$$Effective\ Resolution = \log_2(2^N/RMS\ Input\ Noise)$$

and is expressed in bits.

Total Harmonic Distortion (THD)

THD is the ratio of the rms sum of the first five harmonic components to the rms value of a full-scale input signal and is expressed in dB.

Dynamic Range

Dynamic range is the ratio of the rms value of the full scale to the total rms noise measured with the inputs shorted together. The value for dynamic range is expressed in dB. It is measured with a signal at -60 dBFS to include all noise sources and DNL artifacts.

Signal-to-Noise Ratio (SNR)

SNR is the ratio of the rms value of the actual input signal to the rms sum of all other spectral components below the Nyquist frequency, excluding harmonics and dc. The value for SNR is expressed in dB.

Signal-to-Noise-and-Distortion Ratio (SINAD)

SINAD is the ratio of the rms value of the actual input signal to the rms sum of all other spectral components below the Nyquist frequency, including harmonics but excluding dc. The value for SINAD is expressed in dB.

Aperture Delay

Aperture delay is the measure of the acquisition performance. It is the time between the rising edge of the CNV input and when the input signal is held for a conversion.

Transient Response

Transient response is the time required for the ADC to accurately acquire its input after a full-scale step function is applied.

TYPICAL PERFORMANCE CHARACTERISTICS

$V_{DD} = 2.5\text{ V}$, $V_{REF} = 5.0\text{ V}$, $V_{IO} = 3.3\text{ V}$, unless otherwise noted.

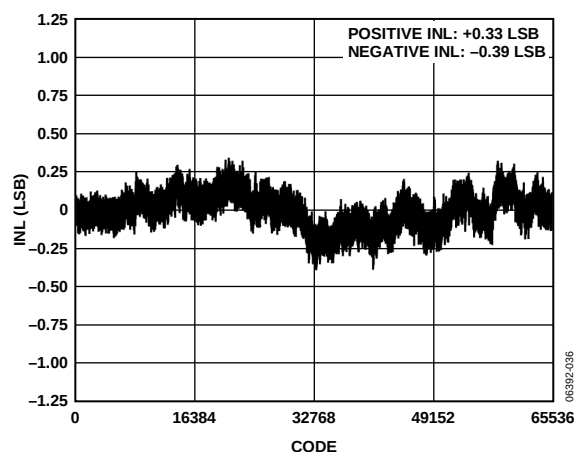


Figure 5. Integral Nonlinearity vs. Code, REF = 5 V

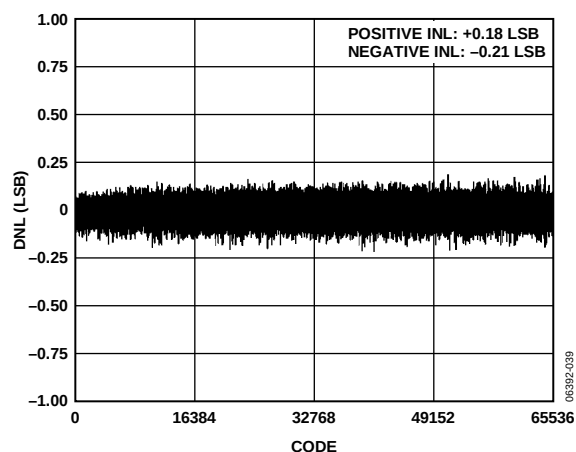


Figure 8. Differential Nonlinearity vs. Code, REF = 5 V

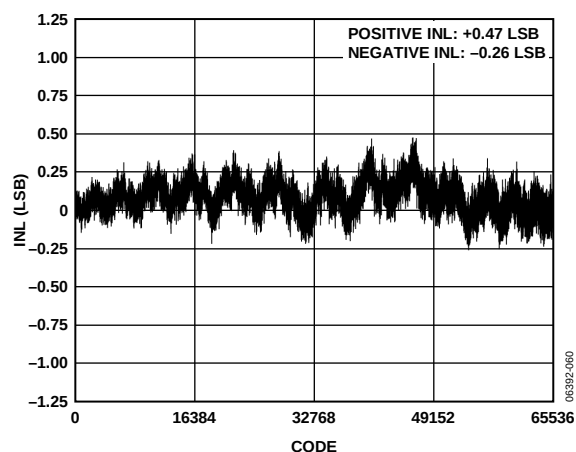


Figure 6. Integral Nonlinearity vs. Code, REF = 2.5 V

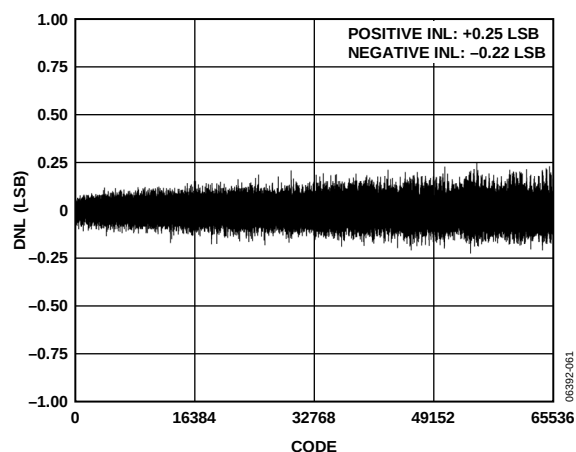


Figure 9. Differential Nonlinearity vs. Code, REF = 2.5 V

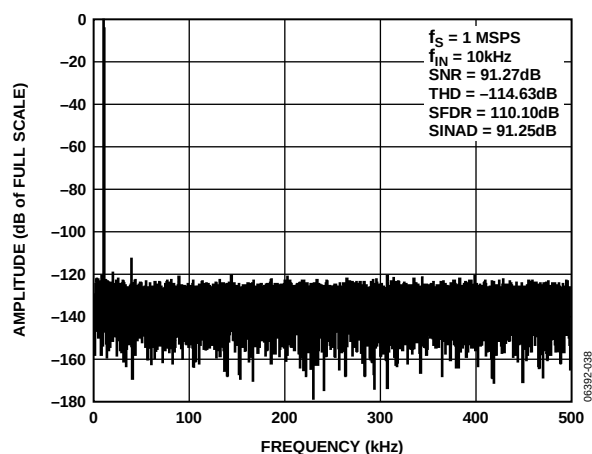


Figure 7. FFT Plot, REF = 5 V

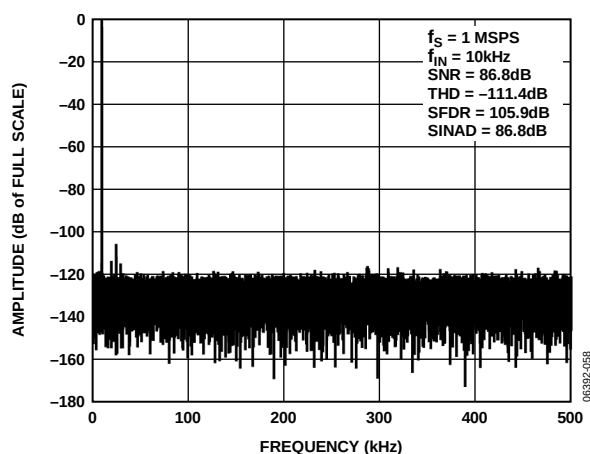


Figure 10. FFT Plot, REF = 2.5 V

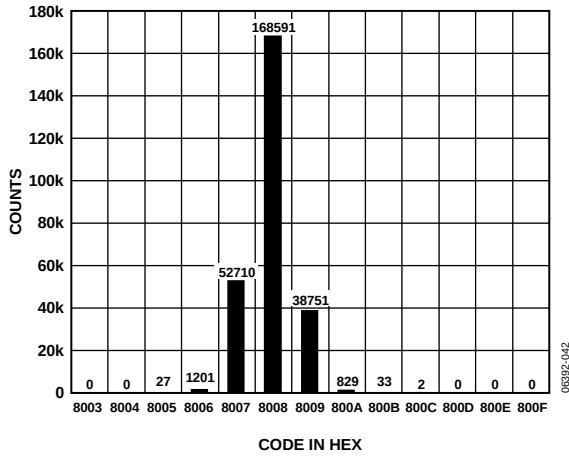


Figure 11. Histogram of a DC Input at the Code Center, REF = 5 V

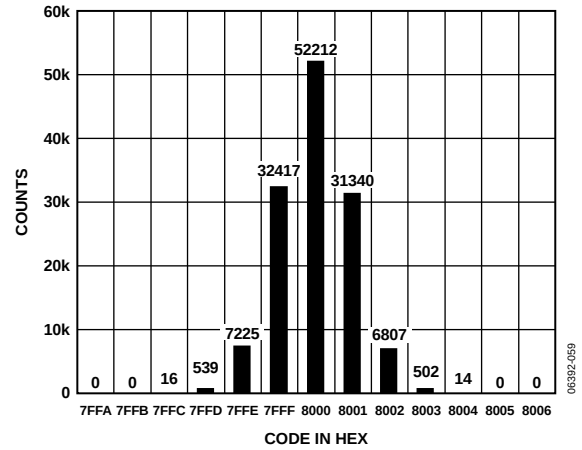


Figure 14. Histogram of a DC Input at the Code Center, REF = 2.5 V

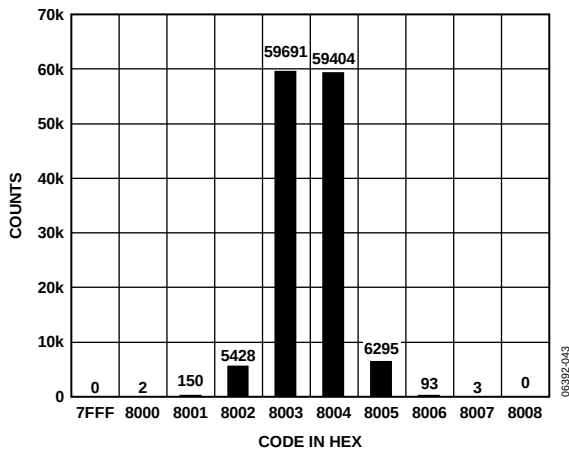


Figure 12. Histogram of a DC Input at the Code Transition, REF = 5 V

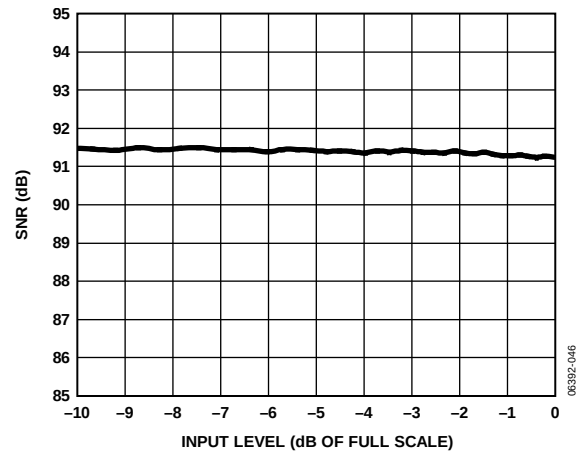


Figure 15. SNR vs. Input Level

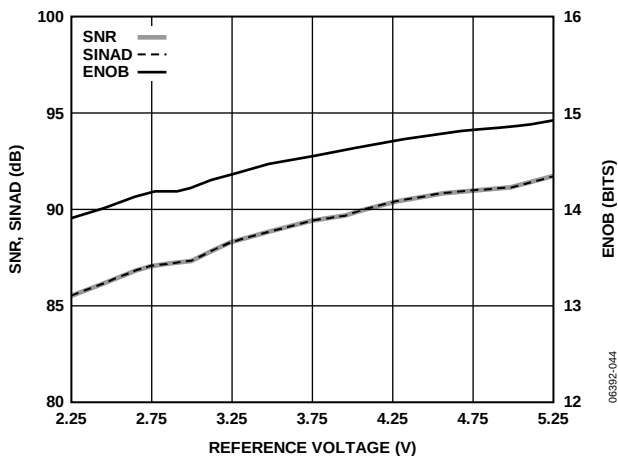


Figure 13. SNR, SINAD, and ENOB vs. Reference Voltage

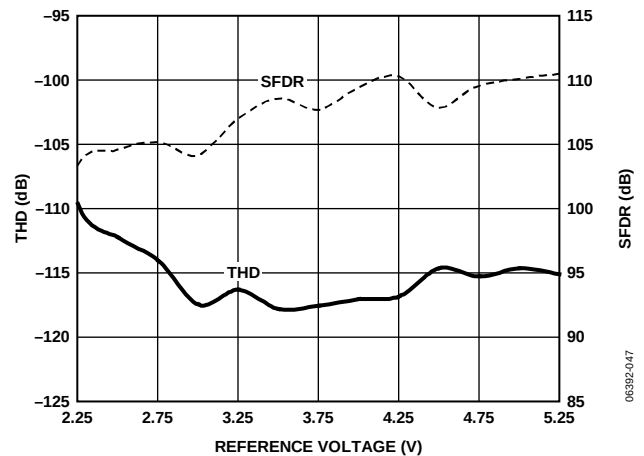


Figure 16. THD, SFDR vs. Reference Voltage

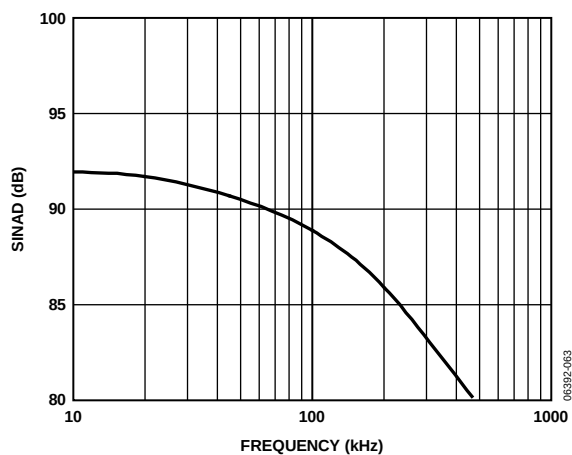


Figure 17. SINAD vs. Frequency

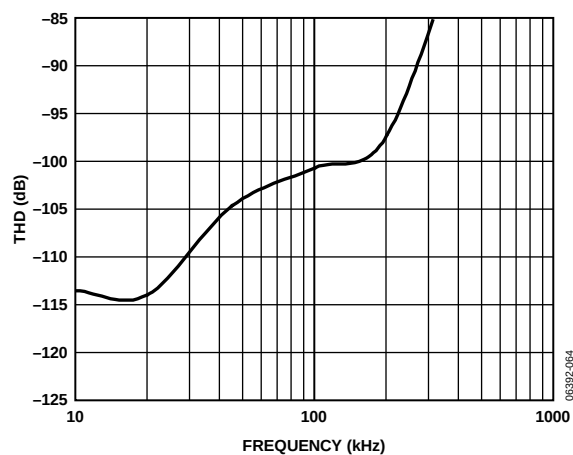


Figure 20. THD vs. Frequency

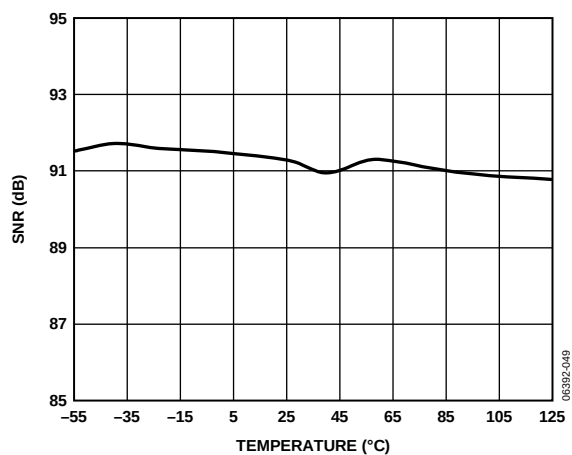


Figure 18. SNR vs. Temperature

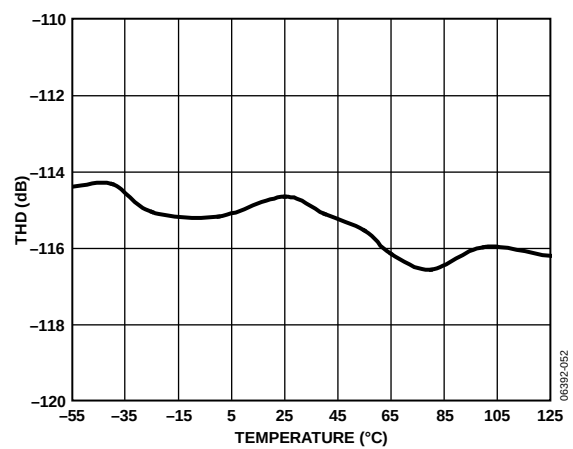


Figure 21. THD vs. Temperature

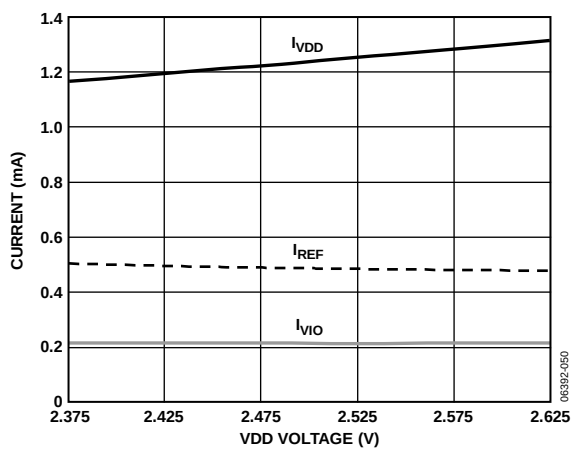


Figure 19. Operating Currents vs. Supply

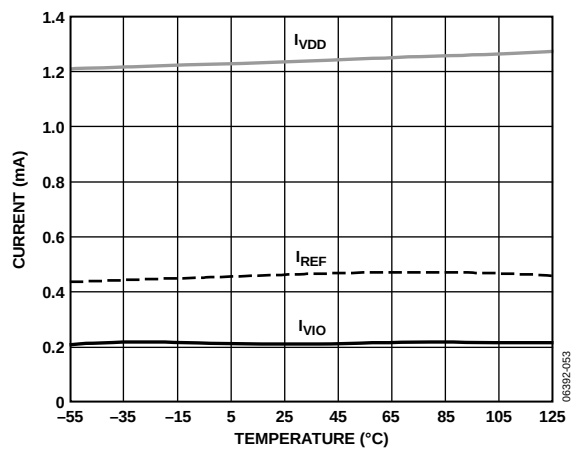


Figure 22. Operating Currents vs. Temperature

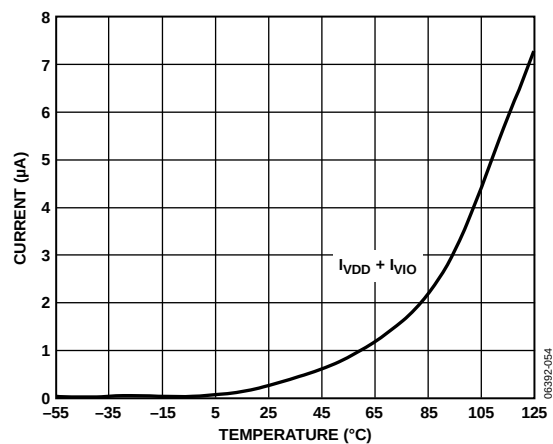


Figure 23. Power-Down Currents vs. Temperature

THEORY OF OPERATION

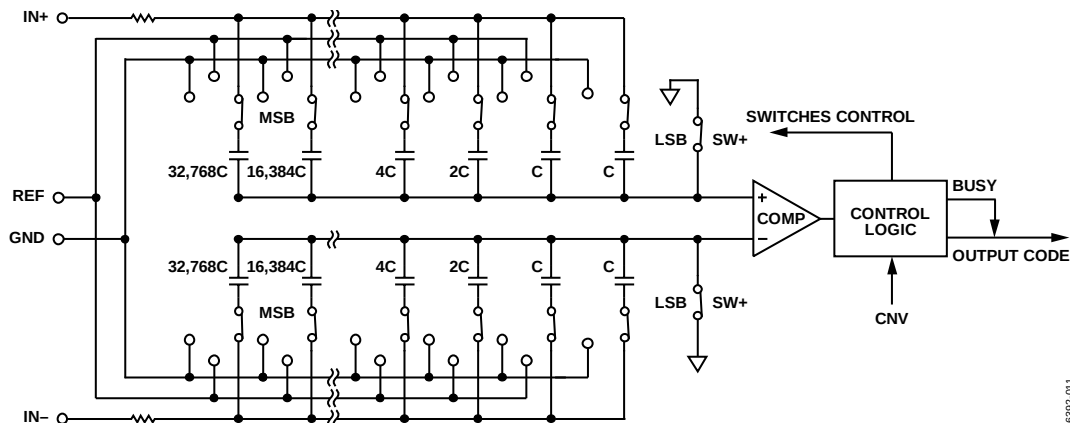


Figure 24. ADC Simplified Schematic

06392-011

CIRCUIT INFORMATION

The [AD7980](#) is a fast, low power, single-supply, precise 16-bit ADC that uses a successive approximation architecture.

The [AD7980](#) is capable of converting 1,000,000 samples per second (1 MSPS) and powers down between conversions. When operating at 10 kSPS, for example, it consumes 70 μ W typically, ideal for battery-powered applications.

The [AD7980](#) provides the user with on-chip track-and-hold and does not exhibit any pipeline delay or latency, making it ideal for multiple multiplexed channel applications.

The [AD7980](#) can be interfaced to any 1.8 V to 5 V digital logic family. It is housed in a 10-lead MSOP or a tiny 10-lead LFCSP that combines space savings and allows flexible configurations.

It is pin-for-pin compatible with the 18-bit [AD7982](#).

CONVERTER OPERATION

The [AD7980](#) is a successive approximation ADC based on a charge redistribution DAC. Figure 24 shows the simplified schematic of the ADC. The capacitive DAC consists of two identical arrays of 16 binary weighted capacitors, which are connected to the two comparator inputs.

During the acquisition phase, terminals of the array tied to the input of the comparator are connected to GND via SW+ and SW-. All independent switches are connected to the analog inputs. Therefore, the capacitor arrays are used as sampling capacitors and acquire the analog signal on the IN+ and IN- inputs. When the acquisition phase is completed and the CNV input goes high, a conversion phase is initiated. When the conversion phase begins, SW+ and SW- are opened first. The two capacitor arrays are then disconnected from the inputs and connected to the GND input. Therefore, the differential voltage between the inputs IN+ and IN- captured at the end of the acquisition phase are applied to the comparator inputs, causing the comparator to become unbalanced. By switching each element of the capacitor array between GND and REF, the comparator input varies by binary weighted voltage steps ($V_{REF}/2$, $V_{REF}/4$... $V_{REF}/65,536$). The control logic toggles these switches, starting with the MSB, to bring the comparator back into a balanced condition. After the completion of this process, the device returns to the acquisition phase and the control logic generates the ADC output code and a busy signal indicator.

Because the [AD7980](#) has an on-board conversion clock, the serial clock, SCK, is not required for the conversion process.

Transfer Functions

The ideal transfer characteristic for the AD7980 is shown in Figure 25 and Table 9.

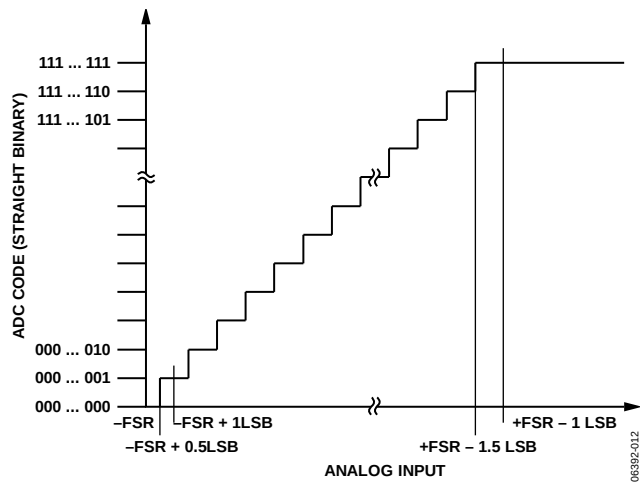


Figure 25. ADC Ideal Transfer Function

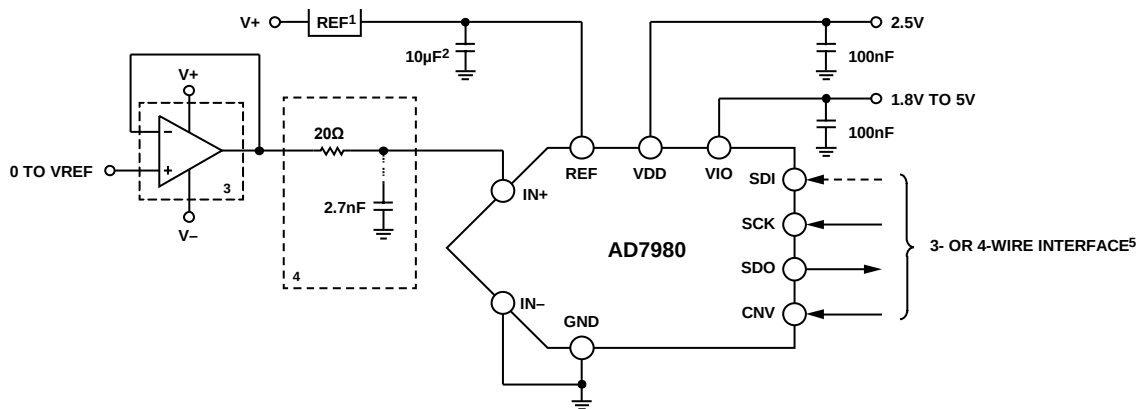
Table 9. Output Codes and Ideal Input Voltages

Description	Analog Input	
	V _{REF} = 5 V	Digital Output Code (Hex)
FSR – 1 LSB	4.999924 V	FFFF ¹
Midscale + 1 LSB	2.500076 V	8001
Midscale	2.5 V	8000
Midscale – 1 LSB	2.499924 V	7FFF
–FSR + 1 LSB	76.3 μV	0001
–FSR	0 V	0000 ²

¹ This is also the code for an overranged analog input ($V_{IN+} - V_{IN-}$ above $V_{REF} - V_{GND}$).
² This is also the code for an underranged analog input ($V_{IN+} - V_{IN-}$ below V_{GND}).

TYPICAL APPLICATION CIRCUIT WITH MULTIPLE SUPPLIES

Figure 26 shows an example of a typical application circuit for the AD7980 when multiple supplies are available.



¹SEE THE VOLTAGE REFERENCE INPUT SECTION FOR REFERENCE SELECTION.
²C_{REF} IS USUALLY A 10μF CERAMIC CAPACITOR (X5R).
³SEE THE DRIVER AMPLIFIER CHOICE SECTION.
⁴RECOMMENDED FILTER CONFIGURATION. SEE THE ANALOG INPUTS SECTION.
⁵SEE THE DIGITAL INTERFACE FOR THE MOST CONVENIENT INTERFACE MODE.

Figure 26. Typical Application Circuit with Multiple Supplies

ANALOG INPUT

Figure 27 shows an equivalent circuit of the input structure of the AD7980.

The two diodes, D1 and D2, provide ESD protection for the analog inputs, IN+ and IN-. Care must be taken to ensure that the analog input signal never exceeds the supply rails by more than 0.3 V, because this causes these diodes to become forward-biased and start conducting current. These diodes can handle a forward-biased current of 130 mA maximum. For instance, these conditions could eventually occur when the supplies of the input buffer (U1) are different from VDD. In such a case (for example, an input buffer with a short circuit), the current limitation can be used to protect the device.

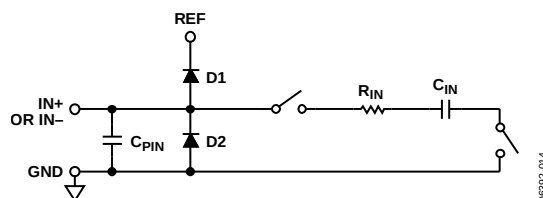


Figure 27. Equivalent Analog Input Circuit

The analog input structure allows the sampling of the true differential signal between IN+ and IN-. By using these differential inputs, signals common to both inputs are rejected.

During the acquisition phase, the impedance of the analog inputs (IN+ and IN-) can be modeled as a parallel combination of capacitor, C_{PIN} , and the network formed by the series connection of R_{IN} and C_{IN} . C_{PIN} is primarily the pin capacitance. R_{IN} is typically 400 Ω and is a lumped component made up of some serial resistors and the on resistance of the switches. C_{IN} is typically 30 pF and is mainly the ADC sampling capacitor. During the conversion phase, where the switches are opened, the input impedance is limited to C_{PIN} . R_{IN} and C_{IN} make a 1-pole, low-pass filter that reduces undesirable aliasing effects and limits the noise.

When the source impedance of the driving circuit is low, the AD7980 can be driven directly. Large source impedances significantly affect the ac performance, especially THD. The dc performances are less sensitive to the input impedance. The maximum source impedance depends on the amount of THD that can be tolerated. The THD degrades as a function of the source impedance and the maximum input frequency.

DRIVER AMPLIFIER CHOICE

Although the AD7980 is easy to drive, the driver amplifier needs to meet the following requirements:

- The noise generated by the driver amplifier needs to be kept as low as possible to preserve the SNR and transition noise performance of the AD7980. The noise coming from the driver is filtered by the 1-pole, low-pass filter of the AD7980 analog input circuit made by R_{IN} and C_{IN} or by the external filter, if one is used. Because the typical noise of the AD7980 is 47.3 μ V rms, the SNR degradation due to the amplifier is

$$SNR_{LOSS} = 20 \log \left(\frac{47.3}{\sqrt{47.3^2 + \frac{\pi}{2} f_{-3dB} (Ne_N)^2}} \right)$$

where:

f_{-3dB} is the input bandwidth in MHz of the AD7980 (10 MHz) or the cutoff frequency of the input filter, if one is used.

N is the noise gain of the amplifier (for example, 1 in buffer configuration).

e_N is the equivalent input noise voltage of the op amp, in nV/ $\sqrt{\text{Hz}}$.

- For ac applications, the driver should have a THD performance commensurate with the AD7980.
- For multichannel multiplexed applications, the driver amplifier and the AD7980 analog input circuit must settle for a full-scale step onto the capacitor array at a 16-bit level (0.0015%, 15 ppm). In the amplifier data sheet, settling at 0.1% to 0.01% is more commonly specified. This can differ significantly from the settling time at a 16-bit level and should be verified prior to driver selection.

The Precision ADC Driver Tool can be used to model the settling behavior and to estimate the ac performance of the AD7980 with a selected driver and RC filter.

Table 10. Recommended Driver Amplifiers¹

Amplifier	Typical Application
ADA4805-1	Low noise, small size, and low power
ADA4807-1	Very low noise and high frequency
ADA4627-1	Precision, low noise, and low input bias current
ADA4522-1	Precision, zero drift, and EMI enhanced
ADA4841-1	Low noise, low distortion, and low power

¹ For the latest recommended drivers, see the product recommendations listed on the product webpage.

VOLTAGE REFERENCE INPUT

The AD7980 voltage reference input, REF, has a dynamic input impedance and should therefore be driven by a low impedance source with efficient decoupling between the REF and GND pins, as explained in the Layout section.

When REF is driven by a very low impedance source, for example, a reference buffer using the AD8031, the ADA4805-1, or the ADA4807-1, a ceramic chip capacitor is appropriate for optimum performance.

If an unbuffered reference voltage is used, the decoupling value depends on the reference used. For instance, a 22 μF (X5R, 1206 size) ceramic chip capacitor is appropriate for optimum performance using a low temperature drift reference, such as the ADR435, ADR445, LTC6655, or ADR4550.

If desired, a reference-decoupling capacitor value as small as 2.2 μF can be used with a minimal impact on performance, especially DNL.

Regardless, there is no need for an additional lower value ceramic decoupling capacitor (for example, 100 nF) between the REF and GND pins.

POWER SUPPLY

The AD7980 uses two power supply pins: a core supply, VDD, and a digital input/output interface supply, VIO. VIO allows direct interface with any logic between 1.8 V and 5.0 V. To reduce the number of supplies needed, VIO and VDD can be tied together. When VIO is greater than or equal to VDD, the AD7980 is insensitive to power supply sequencing. In normal operation, if the magnitude of VIO is less than the magnitude of VDD, VIO must be applied before VDD. Additionally, it is very insensitive to power supply variations over a wide frequency range, as shown in Figure 28.

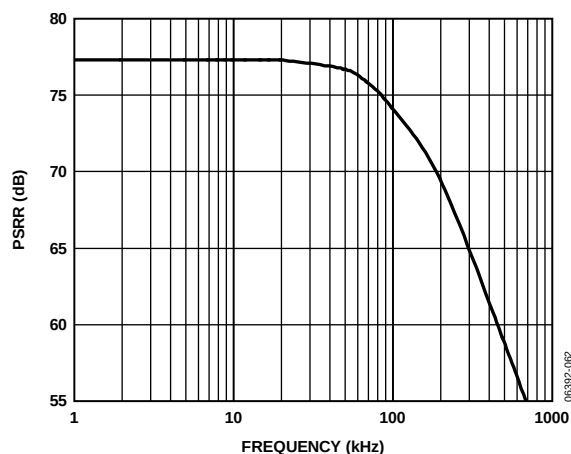


Figure 28. PSRR vs. Frequency

The AD7980 powers down automatically at the end of each conversion phase and, therefore, the power scales linearly with the sampling rate. This makes the device ideal for low sampling rate (even of a few Hz) and low battery-powered applications.

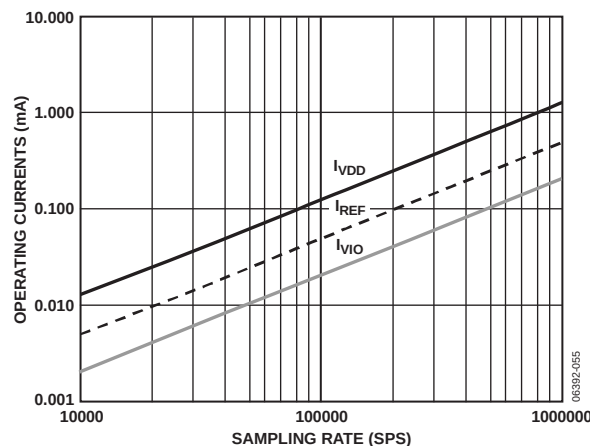


Figure 29. Operating Currents vs. Sampling Rate

DIGITAL INTERFACE

Though the AD7980 has a reduced number of pins, it offers flexibility in its serial interface modes.

The AD7980, when in $\overline{\text{CS}}$ mode, is compatible with SPI, QSPI™, and digital hosts. This interface can use either a 3-wire or 4-wire interface. A 3-wire interface using the CNV, SCK, and SDO signals minimizes wiring connections useful, for instance, in isolated applications. A 4-wire interface using the SDI, CNV, SCK, and SDO signals allows CNV, which initiates the conversions, to be independent of the readback timing (SDI). This is useful in low jitter sampling or simultaneous sampling applications.

The AD7980, when in chain mode, provides a daisy-chain feature using the SDI input for cascading multiple ADCs on a single data line similar to a shift register.

The mode in which the device operates depends on the SDI level when the CNV rising edge occurs. The $\overline{\text{CS}}$ mode is selected if SDI is high, and the chain mode is selected if SDI is low. The SDI hold time is such that when SDI and CNV are connected together, the chain mode is selected.

In either mode, the AD7980 offers the flexibility to optionally force a start bit in front of the data bits. This start bit can be used as a busy signal indicator to interrupt the digital host and trigger the data reading. Otherwise, without a busy indicator, the user must time out the maximum conversion time prior to readback.

The busy indicator feature is enabled in the $\overline{\text{CS}}$ mode if CNV or SDI is low when the ADC conversion ends (see Figure 33 and Figure 37). The busy indicator feature is enabled in the chain mode if SCK is high during the CNV rising edge (see Figure 41).

3-WIRE $\overline{\text{CS}}$ MODE WITHOUT BUSY INDICATOR

This mode is usually used when a single AD7980 is connected to an SPI-compatible digital host. The connection diagram is shown in Figure 30, and the corresponding timing is given in Figure 31.

With SDI tied to VIO, a rising edge on CNV initiates a conversion, selects the $\overline{\text{CS}}$ mode, and forces SDO to high impedance. Once a conversion is initiated, it continues until completion irrespective of the state of CNV. This can be useful, for instance, to bring CNV low to select other SPI devices, such as analog multiplexers; however, CNV must be returned high before the minimum conversion time elapses and then held high for the maximum conversion time to avoid the generation of the busy signal indicator. When the conversion is complete, the AD7980 enters the acquisition phase and powers down.

When CNV goes low, the MSB is output onto SDO. The remaining data bits are then clocked by subsequent SCK falling edges. The data is valid on both SCK edges. Although the rising edge can be used to capture the data, a digital host using the SCK falling edge allows a faster reading rate provided that it has an acceptable hold time. After the 16th SCK falling edge or when CNV goes high, whichever is earlier, SDO returns to high impedance.

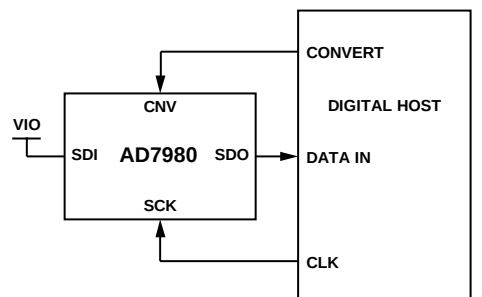


Figure 30. 3-Wire $\overline{\text{CS}}$ Mode Without Busy Indicator Connection Diagram (SDI High)

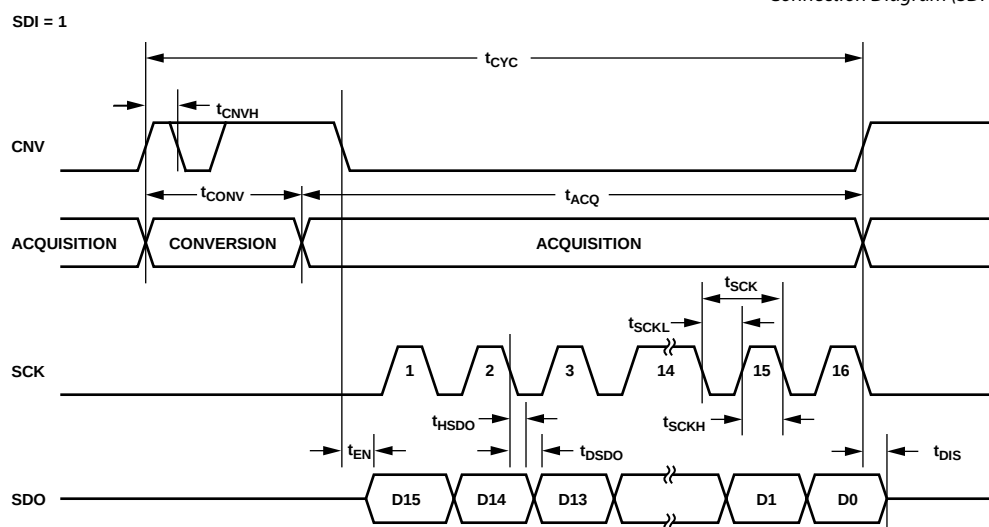


Figure 31. 3-Wire $\overline{\text{CS}}$ Mode Without Busy Indicator Serial Interface Timing (SDI High)

3-WIRE $\overline{\text{CS}}$ MODE WITH BUSY INDICATOR

This mode is usually used when a single AD7980 is connected to an SPI-compatible digital host having an interrupt input.

The connection diagram is shown in Figure 32, and the corresponding timing is given in Figure 33.

With SDI tied to VIO, a rising edge on CNV initiates a conversion, selects the $\overline{\text{CS}}$ mode, and forces SDO to high impedance. SDO is maintained in high impedance until the completion of the conversion irrespective of the state of CNV. Prior to the minimum conversion time, CNV can be used to select other SPI devices, such as analog multiplexers, but CNV must be returned low before the minimum conversion time elapses and then held low for the maximum conversion time to guarantee the generation of the busy signal indicator. When the conversion is complete, SDO goes from high impedance to low. With a pull-up on the SDO line, this transition can be used as an interrupt signal to initiate the data reading controlled by the digital host. The AD7980 then enters the acquisition phase and powers down. The data bits are clocked out, MSB first, by subsequent SCK falling edges. The data is valid on both SCK edges. Although the rising edge can be used to capture the data, a digital host using the SCK falling edge allows a faster reading rate provided it has an acceptable hold time. After the optional 17th SCK falling edge or when CNV goes high, whichever is earlier, SDO returns to high impedance.

If multiple AD7980 devices are selected at the same time, the SDO output pin handles this contention without damage or induced latch-up. Meanwhile, it is recommended to keep this contention as short as possible to limit extra power dissipation.

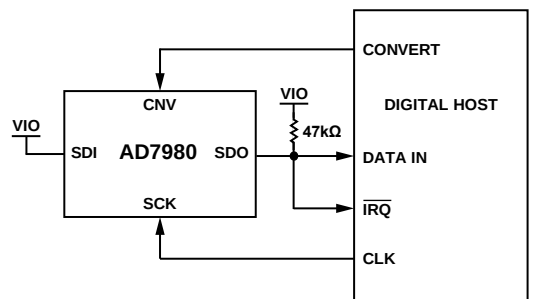


Figure 32. 3-Wire $\overline{\text{CS}}$ Mode with Busy Indicator Connection Diagram (SDI High)

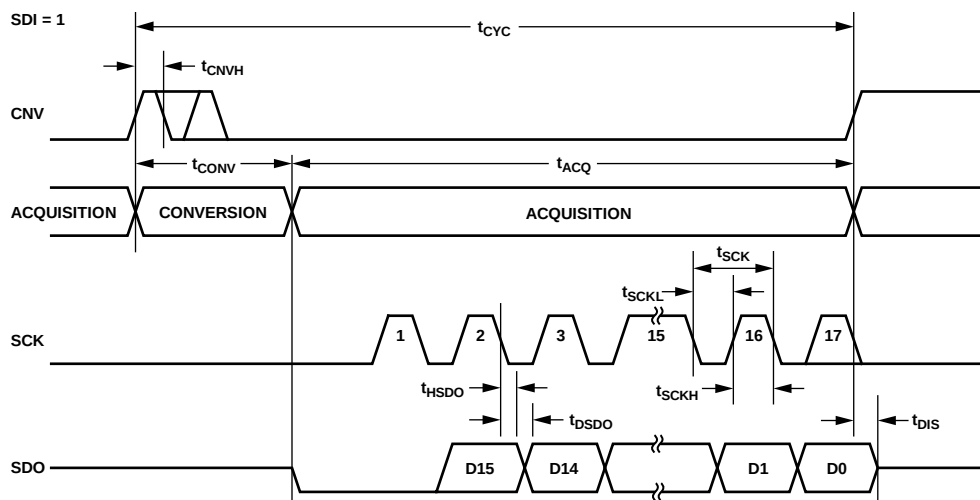


Figure 33. 3-Wire $\overline{\text{CS}}$ Mode with Busy Indicator Serial Interface Timing (SDI High)

With SDI high, a rising edge on CNV initiates a conversion, selects the CS mode, and forces SDO to high impedance. In this mode, CNV must be held high during the conversion phase and the subsequent data readback (if SDI and CNV are low, SDO is driven low). Prior to the minimum conversion time, SDI can be used to select other SPI devices, such as analog multiplexers, but SDI must be returned high before the minimum conversion time elapses and then held high for the maximum conversion time to avoid the generation of the busy signal indicator.

When the conversion is complete, the [AD7980](#) enters the acquisition phase and powers down. Each ADC result can be read by bringing its SDI input low, which consequently outputs the MSB onto SDO. The remaining data bits are then clocked by subsequent SCK falling edges. The data is valid on both SCK edges. Although the rising edge can be used to capture the data, a digital host using the SCK falling edge allows a faster reading rate provided it has an acceptable hold time. After the 16th SCK falling edge or when SDI goes high, whichever is earlier, SDO returns to high impedance and another [AD7980](#) can be read.



4-WIRE $\overline{\text{CS}}$ MODE WITH BUSY INDICATOR

This mode is usually used when a single AD7980 is connected to an SPI-compatible digital host that has an interrupt input, and it is desired to keep CNV, which is used to sample the analog input, independent of the signal used to select the data reading. This requirement is particularly important in applications where low jitter on CNV is desired.

The connection diagram is shown in Figure 36, and the corresponding timing is given in Figure 37.

With SDI high, a rising edge on CNV initiates a conversion, selects the $\overline{\text{CS}}$ mode, and forces SDO to high impedance. In this mode, CNV must be held high during the conversion phase and the subsequent data readback (if SDI and CNV are low, SDO is driven low). Prior to the minimum conversion time, SDI can be used to select other SPI devices, such as analog multiplexers, but SDI must be returned low before the minimum conversion time elapses and then held low for the maximum conversion time to guarantee the generation of the busy signal indicator. When the conversion is complete, SDO goes from high impedance to low.

With a pull-up on the SDO line, this transition can be used as an interrupt signal to initiate the data readback controlled by the digital host. The AD7980 then enters the acquisition phase and powers down. The data bits are clocked out, MSB first, by subsequent SCK falling edges. The data is valid on both SCK edges. Although the rising edge can be used to capture the data, a digital host using the SCK falling edge allows a faster reading rate provided it has an acceptable hold time. After the optional 17th SCK falling edge or SDI going high, whichever is earlier, the SDO returns to high impedance.

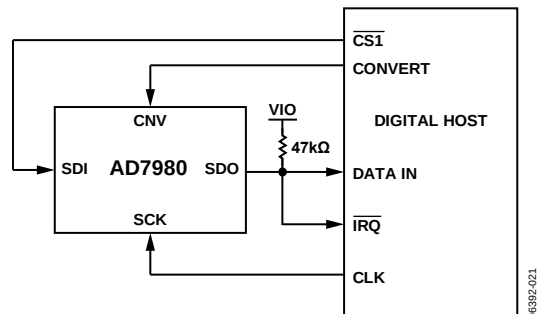


Figure 36. 4-Wire $\overline{\text{CS}}$ Mode with Busy Indicator Connection Diagram

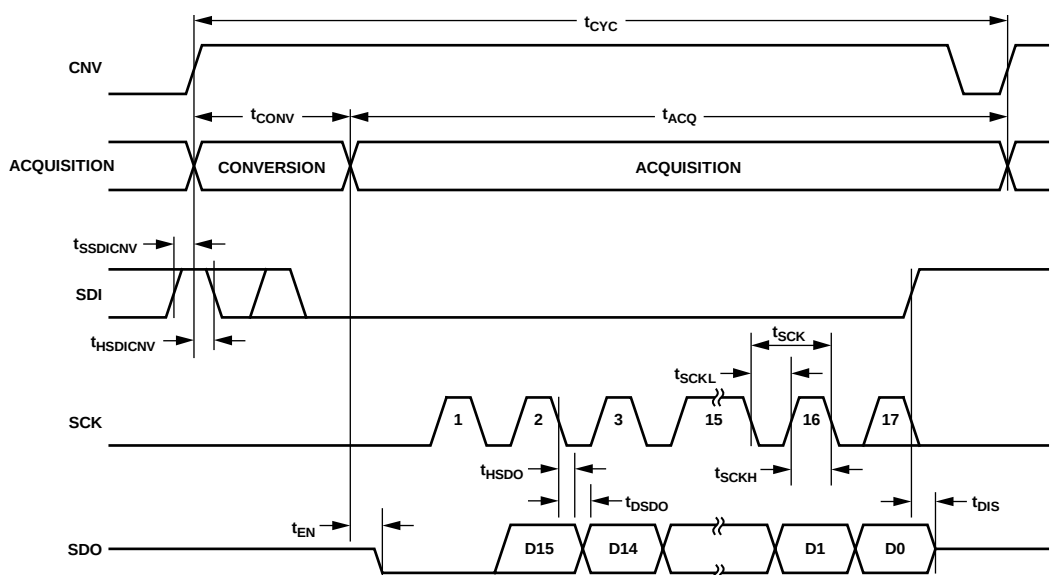


Figure 37. 4-Wire $\overline{\text{CS}}$ Mode with Busy Indicator Serial Interface Timing

CHAIN MODE WITHOUT BUSY INDICATOR

This mode can be used to daisy-chain multiple AD7980 devices on a 3-wire serial interface. This feature is useful for reducing component count and wiring connections, for example, in isolated multi-converter applications or for systems with a limited interfacing capacity. Data readback is analogous to clocking a shift register.

A connection diagram example using two AD7980s is shown in Figure 38, and the corresponding timing is given in Figure 39.

When SDI and CNV are low, SDO is driven low. With SCK low, a rising edge on CNV initiates a conversion, selects the chain mode, and disables the busy indicator. In this mode, CNV is held high during the conversion phase and the

subsequent data readback. When the conversion is complete, the MSB is output onto SDO and the AD7980 enters the acquisition phase and powers down. The remaining data bits stored in the internal shift register are clocked by subsequent SCK falling edges. For each ADC, SDI feeds the input of the internal shift register and is clocked by the SCK falling edge. Each ADC in the chain outputs its data MSB first, and $16 \times N$ clocks are required to readback the N ADCs. The data is valid on both SCK edges. Although the rising edge can be used to capture the data, a digital host using the SCK falling edge allows a faster reading rate and, consequently, more AD7980 devices in the chain, provided the digital host has an acceptable hold time. The maximum conversion rate may be reduced due to the total readback time.

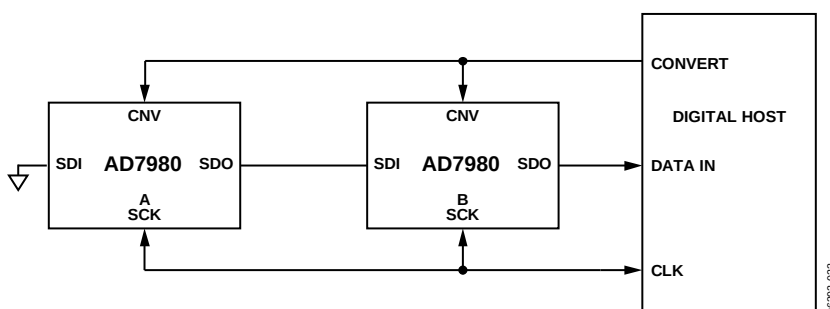


Figure 38. Chain Mode Without Busy Indicator Connection Diagram

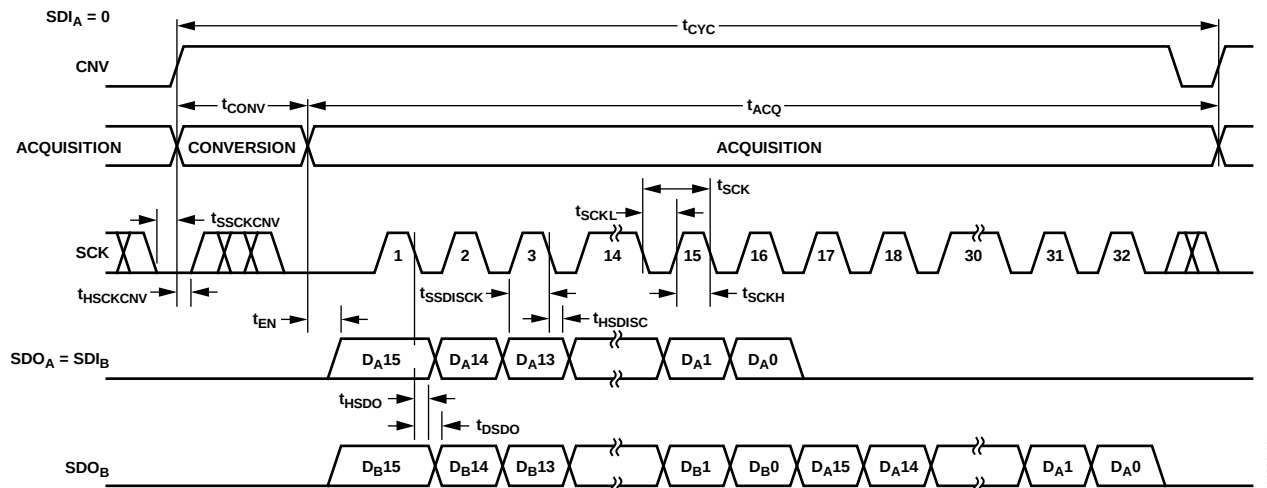
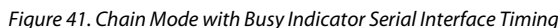
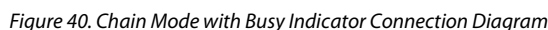


Figure 39. Chain Mode Without Busy Indicator Serial Interface Timing

This mode can also be used to daisy-chain multiple [AD7980](#) devices on a 3-wire serial interface while providing a busy indicator. This feature is useful for reducing component count and wiring connections, for example, in isolated multiconverter applications or for systems with a limited interfacing capacity. Data readback is analogous to clocking a shift register.

When SDI and CNV are low, SDO is driven low. With SCK high, a rising edge on CNV initiates a conversion, selects the chain mode, and enables the busy indicator feature. In this mode, CNV is held high during the conversion phase and the

subsequent data readback. When all ADCs in the chain have completed their conversions, the SDO pin of the ADC closest to the digital host (see the [AD7980](#) ADC labeled C in Figure 40) is driven high. This transition on SDO can be used as a busy indicator to trigger the data readback controlled by the digital host. The [AD7980](#) then enters the acquisition phase and powers down. The data bits stored in the internal shift register are clocked out, MSB first, by subsequent SCK falling edges. For each ADC, SDI feeds the input of the internal shift register and is clocked by the SCK falling edge. Each ADC in the chain outputs its data MSB first, and $16 \times N + 1$ clocks are required to readback the N ADCs. Although the rising edge can be used to capture the data, a digital host using the SCK falling edge allows a faster reading rate and, consequently, more [AD7980](#) devices in the chain, provided the digital host has an acceptable hold time.



APPLICATIONS INFORMATION

LAYOUT

The printed circuit board (PCB) that houses the [AD7980](#) should be designed so that the analog and digital sections are separated and confined to certain areas of the board. The pinout of the [AD7980](#), with all its analog signals on the left side and all its digital signals on the right side, eases this task.

Avoid running digital lines under the device because these couple noise onto the die, unless a ground plane under the [AD7980](#) is used as a shield. Fast switching signals, such as CNV or clocks, should never run near analog signal paths. Crossover of digital and analog signals should be avoided.

At least one ground plane should be used. The ground plane can be common or split between the digital and analog section. In the latter case, the planes should be joined underneath the [AD7980](#) devices.

The [AD7980](#) voltage reference input REF has a dynamic input impedance and should be decoupled with minimal parasitic inductances. This is done by placing the reference decoupling ceramic capacitor close to, ideally right up against, the REF and GND pins and connecting them with wide, low impedance traces.

Finally, the power supplies VDD and VIO of the [AD7980](#) should be decoupled with ceramic capacitors, typically 100 nF, placed close to the [AD7980](#) and connected using short and wide traces to provide low impedance paths and reduce the effect of glitches on the power supply lines.

An example of a layout following these rules is shown in Figure 42 and Figure 43.

EVALUATING THE PERFORMANCE OF THE [AD7980](#)

Other recommended layouts for the [AD7980](#) are outlined in the documentation of the evaluation board for the [AD7980](#) ([EVAL-AD7980SDZ](#)). The evaluation board package includes a fully assembled and tested evaluation board, documentation, and software for controlling the board from a PC via the [EVAL-SDP-CB1Z](#).

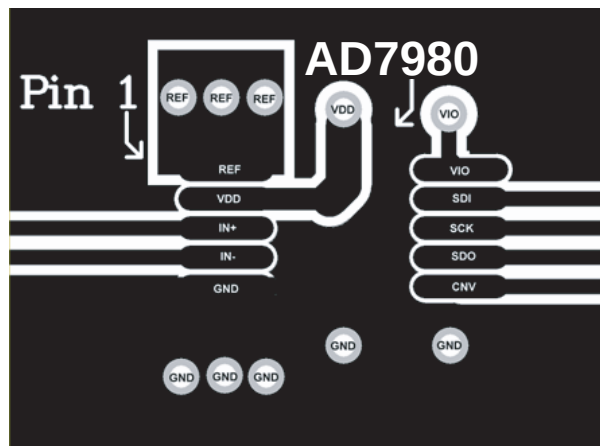


Figure 42. Example Layout of the [AD7980](#) (Top Layer)

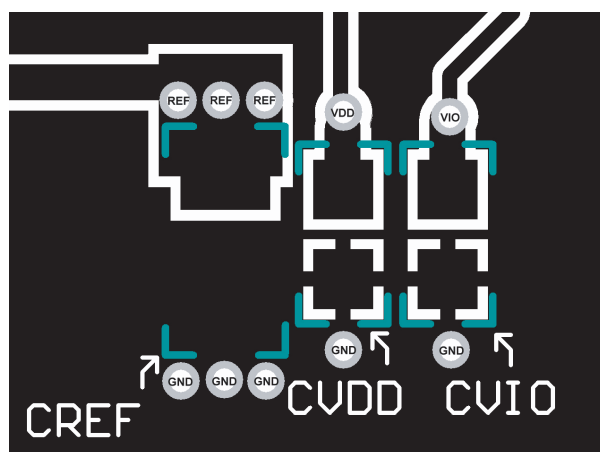
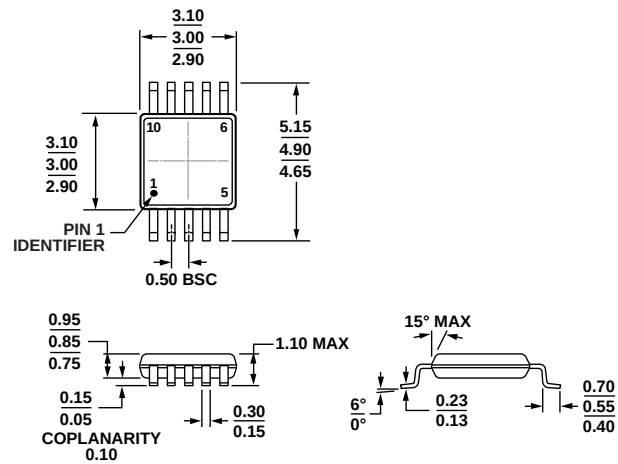


Figure 43. Example Layout of the [AD7980](#) (Bottom Layer)

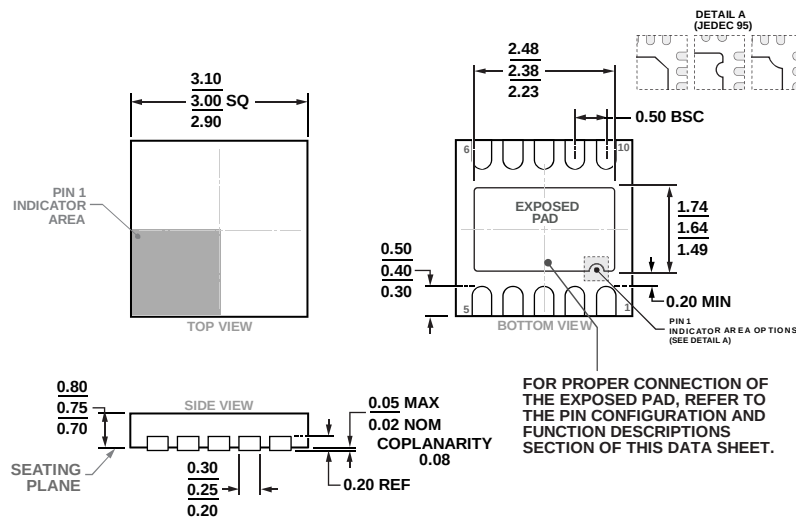
OUTLINE DIMENSIONS



COMPLIANT TO JEDEC STANDARDS MO-187-BA

Figure 44. 10-Lead Mini Small Outline Package [MSOP]
(RM-10)

Dimensions shown in millimeters

Figure 45. 10-Lead Lead Frame Chip Scale Package [LFCSP]
3 mm × 3 mm Body and 0.75 mm Package Height
(CP-10-9)

Dimensions shown in millimeters

Contact sales for the non-RoHS compliant version of the device.

ORDERING GUIDE

Model ^{1, 2, 3}	Integral Nonlinearity	Temperature Range	Package Description	Package Option	Branding	Ordering Quantity
AD7980ARMZ	±2.5 LSB max	−40°C to +125°C	10-Lead MSOP	RM-10	C5X	Tube, 50
AD7980ARMZRL7	±2.5 LSB max	−40°C to +125°C	10-Lead MSOP	RM-10	C5X	Reel, 1,000
AD7980BRMZ	±1.25 LSB max	−40°C to +125°C	10-Lead MSOP	RM-10	C5D	Tube, 50
AD7980BRMZRL7	±1.25 LSB max	−40°C to +125°C	10-Lead MSOP	RM-10	C5D	Reel, 1,000
AD7980ACPZ-RL	±2.5 LSB max	−40°C to +125°C	10-Lead LFCSP	CP-10-9	C5X	Reel, 5,000
AD7980ACPZ-RL7	±2.5 LSB max	−40°C to +125°C	10-Lead LFCSP	CP-10-9	C5X	Reel, 1,000
AD7980BCPZ-RL	±1.25 LSB max	−40°C to +125°C	10-Lead LFCSP	CP-10-9	C5D	Reel, 5,000
AD7980BCPZ-RL7	±1.25 LSB max	−40°C to +125°C	10-Lead LFCSP	CP-10-9	C5D	Reel, 1,000
AD7980BCPZ-R2	±1.25 LSB max	−40°C to +125°C	10-Lead LFCSP	CP-10-9	C5D	Reel, 1,000
EVAL-AD7980SDZ			Evaluation Board			
EVAL-SDP-CB1Z			Controller Board			

¹ Z = RoHS Compliant Part.

² The [EVAL-AD7980SDZ](#) can be used as a standalone evaluation board or in conjunction with the [EVAL-SDP-CB1Z](#) for evaluation/demonstration purposes.

³ The [EVAL-SDP-CB1Z](#) allows a PC to control and communicate with all Analog Devices evaluation boards ending in the SD designator.

